

Q 1

ASCHAM

Consider the rectangle below.

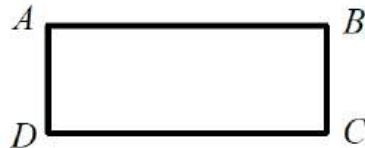
Which of the following statements is false?

A. $\vec{AB} + \vec{BC} = \vec{AC}$

B. $\vec{AB} - \vec{BC} = \vec{DB}$

C. $\vec{AB} - \vec{BC} = \vec{BD}$

D. $\vec{AB} \bullet \vec{BC} = 0$



Q 2

BAULKHAM

If $\underline{u} = 2\hat{i} + 3\hat{j}$, the unit vector in the direction of $\underline{u}$ is:

A. $\frac{1}{13}(2\hat{i} + 3\hat{j})$

C. $\frac{1}{\sqrt{13}}(2\hat{i} + 3\hat{j})$

B. $\frac{1}{5}(2\hat{i} + 3\hat{j})$

D. $\sqrt{13}(2\hat{i} + 3\hat{j})$

$$\begin{aligned}\hat{u} &= \frac{\underline{u}}{|\underline{u}|} \\ &= \frac{2\hat{i} + 3\hat{j}}{\sqrt{2^2 + 3^2}} \\ &= \frac{1}{\sqrt{13}}(2\hat{i} + 3\hat{j})\end{aligned}$$

Q 3

BLACKTOWN

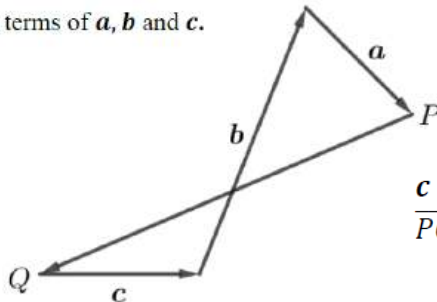
Express the vector $\vec{PQ}$ in terms of $\underline{a}$, $\underline{b}$ and $\underline{c}$.

A. $-\underline{a} - \underline{b} - \underline{c}$

B. $\underline{a} + \underline{b} + \underline{c}$

C. $-\underline{a} + \underline{b} + \underline{c}$

D. $\underline{a} - \underline{b} - \underline{c}$



$$\begin{aligned}\underline{c} + \underline{b} + \underline{a} &= \vec{QP} \\ \vec{PQ} &= -\underline{a} - \underline{b} - \underline{c}\end{aligned}$$

Q 4

CARINGBAH

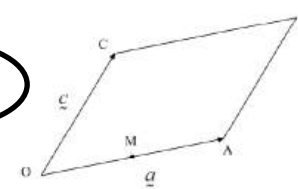
In the diagram, $OABC$ is parallelogram. The vector $\vec{OA} = \underline{a}$ and $\vec{OC} = \underline{c}$. M is the midpoint of OA . Which of the following expressions is represented by the vector $\vec{MB}$?

(A) $\frac{1}{2}\underline{a} - \underline{c}$

(B) $\frac{1}{2}\underline{a} + \underline{c}$

(C) $\underline{a} - \frac{1}{2}\underline{c}$

(D) $\underline{a} + \frac{1}{2}\underline{c}$



Q 5

CSSA

Find the exact magnitude and direction (to the nearest degree) of vector $\vec{AB}$ where

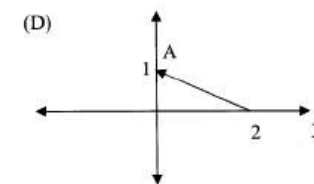
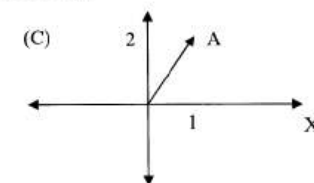
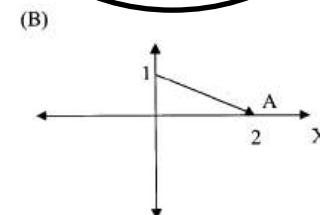
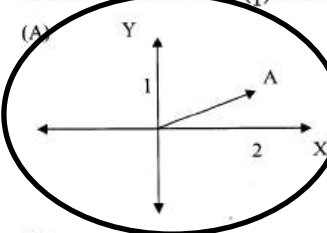
$$\vec{OA} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \text{ and } \vec{OB} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

$$\begin{aligned}|\vec{AB}| &= \sqrt{(3-2)^2 + (-1-1)^2} \\ &= \sqrt{5} \\ \tan \theta &= \frac{-1-1}{3-1} \\ \theta &= \tan^{-1}(-2) \\ &= 297^\circ \text{ (nearest degree)}\end{aligned}$$

Q 6

ENTERPRISE

The position vector $\vec{OA} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is best represented by



vectors in 2 dimensions

Part A. B & C: Introduction

3unit

selective

Q 7

ENTERPRISE

Which of the following is true for the vector $\vec{OB} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$

- (A) magnitude = 5 and direction is 53°
- (B) magnitude = 25 and direction is 37°
- (C) magnitude = 5 and direction is 37°
- (D) magnitude = 7 and direction is 37°

Q 8

ENTERPRISE

If $\vec{OA} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$ and $\vec{OB} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$, find the

- (i) magnitude of $\vec{k} = \vec{OB} - \vec{OA}$ 2
- (ii) direction of $\vec{k} = \vec{OB} - \vec{OA}$ 2

$$(i) \quad \vec{k} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} - \begin{pmatrix} -2 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \end{pmatrix}$$

$$(ii) \quad \tan \theta = \frac{y}{x} = \frac{-3}{6} = -\frac{1}{2}$$

$$\theta = 360^\circ - 27^\circ = 333^\circ \text{ (4th quadrant)}$$

$$|\vec{k}| = \sqrt{x^2 + y^2} = \sqrt{6^2 + (-3)^2} = \sqrt{45} = 6.7$$

Q 9

GIRRAWEEEN

The unit vector in the direction of $\vec{u} = \vec{i} - 2\vec{j}$ is

A. $5(\vec{i} - 2\vec{j})$

B. $\frac{1}{5}(\vec{i} - 2\vec{j})$

C. $\sqrt{5}(\vec{i} - 2\vec{j})$

D. $\frac{1}{\sqrt{5}}(\vec{i} - 2\vec{j})$

unit vector = $\frac{\vec{i} - 2\vec{j}}{\sqrt{1^2 + (-2)^2}} = \frac{\vec{i} - 2\vec{j}}{\sqrt{5}}$

Q10

GRAFTON

The column vector notation of four vectors is shown below.

$$P = \begin{pmatrix} -8 \\ -8 \end{pmatrix} \quad Q = \begin{pmatrix} 3 \\ 6 \end{pmatrix} \quad R = \begin{pmatrix} 1 \\ 5 \end{pmatrix} \quad S = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$$

Find

- (i) $\vec{PQ}$ 1
- (ii) $\vec{RS}$ 1
- (iii) $-\vec{PQ} - \vec{RS}$ 1

Q11) c)

$$i) \quad \vec{PQ} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} - \begin{pmatrix} -8 \\ -8 \end{pmatrix} = \begin{pmatrix} 11 \\ 14 \end{pmatrix} \checkmark$$

$$ii) \quad \vec{RS} = \begin{pmatrix} -5 \\ 7 \end{pmatrix} - \begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} -6 \\ 2 \end{pmatrix} \checkmark$$

$$iii) \quad -\vec{PQ} - \vec{RS} = -\begin{pmatrix} 11 \\ 14 \end{pmatrix} - \begin{pmatrix} -6 \\ 2 \end{pmatrix} = \begin{pmatrix} -5 \\ -16 \end{pmatrix} \checkmark$$

Q11

HUNTERS HILL

If A and B are the points $(-1, 4)$ and $(3, 7)$ respectively, then the unit vector in the direction of $\vec{AB}$ is

(A) $\frac{2}{5}\vec{i} + \frac{11}{5}\vec{j}$

(B) $4\vec{i} + 3\vec{j}$

(C) $-3\vec{i} + 28\vec{j}$

(D) $\frac{4}{5}\vec{i} + \frac{3}{5}\vec{j}$

$$\vec{AB} = \begin{pmatrix} 3 \\ 7 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$|\vec{AB}| = \sqrt{4^2 + 3^2} = 5$$

Q12

JAMES RUSE

How many of the following statements are true?

- $|\underline{a}| + |\underline{b}| = |\underline{a} + \underline{b}|$ means that $\underline{a}$ and $\underline{b}$ have the same direction.
- $|\underline{a}| + |\underline{b}| = |\underline{a} - \underline{b}|$ means that $\underline{a}$ and $\underline{b}$ have the opposite directions.
- $|\underline{a}| + |\underline{b}| = |\underline{a} - \underline{b}|$ means that $\underline{a}$ and $\underline{b}$ have the same magnitude.
- $|\underline{a}| - |\underline{b}| = |\underline{a} - \underline{b}|$ means that $\underline{a}$ and $\underline{b}$ have the same direction.
- (A) 1 (B) 2 (C) 3 (D) 4

Q13

KAMBALA

Given position vectors $\overrightarrow{OA} = -2\mathbf{i} + 4\mathbf{j}$ and $\overrightarrow{OB} = 4\mathbf{i} + 3\mathbf{j}$.

What is the value of $|\overrightarrow{OA} + \overrightarrow{OB}|$?

- (A) 5 (C) $\sqrt{37}$
- (B) $2\sqrt{5}$ (D) $\sqrt{53}$

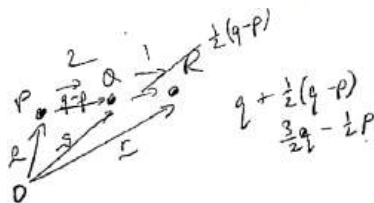
$$\begin{aligned}\overrightarrow{OA} + \overrightarrow{OB} &= 2\mathbf{i} + 7\mathbf{j} \\ |\overrightarrow{OA} + \overrightarrow{OB}| &= \sqrt{2^2 + 7^2} \\ &= \sqrt{4 + 49} = \sqrt{53}\end{aligned}$$

Q14

KAMBALA

P, Q and R are three collinear points with position vectors $\underline{p}, \underline{q}$ and $\underline{r}$ respectively, where Q lies between P and R . If $|\overrightarrow{QR}| = \frac{1}{2} |\overrightarrow{PQ}|$, then $\underline{r}$ is equal to:

- (A) $\frac{3}{2}\underline{q} - \frac{1}{2}\underline{p}$ (C) $\frac{3}{2}\underline{q} - \frac{3}{2}\underline{p}$
- (B) $\frac{3}{2}\underline{p} + \frac{1}{2}\underline{q}$ (D) $\frac{1}{2}\underline{p} - \frac{3}{2}\underline{q}$



Q15

MANLY

Consider the vectors $\underline{a} = 2\mathbf{i} + 3\mathbf{j}$, $\underline{b} = -3\mathbf{i} + 2\mathbf{j}$ and $\underline{c} = 2\mathbf{i} - \mathbf{j}$

Which of the following vectors is parallel to $\underline{a} + \underline{b} + \underline{c}$?

- A. $2\mathbf{i} + 8\mathbf{j}$ C. $2\mathbf{i} - 6\mathbf{j}$ $\underline{a} = 2\mathbf{i} + 3\mathbf{j}$, $\underline{b} = -3\mathbf{i} + 2\mathbf{j}$ and $\underline{c} = 2\mathbf{i} - \mathbf{j}$
- B. $2\mathbf{i} - 8\mathbf{j}$ D. $-2\mathbf{i} - 6\mathbf{j}$ $\underline{a} + \underline{b} + \underline{c} = 2\mathbf{i} + 3\mathbf{j} - 3\mathbf{i} + 2\mathbf{j} + 2\mathbf{i} - \mathbf{j} = \mathbf{i} + 4\mathbf{j}$

Parallel
 $\underline{k}(\mathbf{i} + 4\mathbf{j})$
eg
 $2(\mathbf{i} + 4\mathbf{j})$

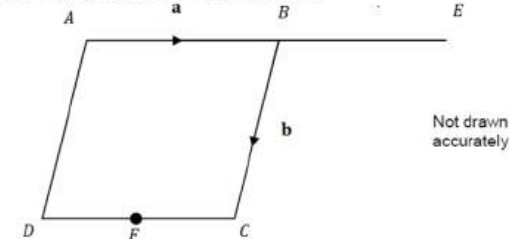
Q16

MANLY

$ABCD$ is a rhombus. $\overrightarrow{BE}$ is an extension of $\overrightarrow{AB}$ such that

$$|\overrightarrow{AB}| : |\overrightarrow{BE}| = 4 : 3$$

$$\begin{aligned}\overrightarrow{AB} &= \underline{a} \\ \overrightarrow{BC} &= \underline{b}\end{aligned}$$



The point F is halfway along CD .

Find $\overrightarrow{FE}$ in terms of $\underline{a}$ and $\underline{b}$

$$\begin{aligned}\overrightarrow{DE} - \overrightarrow{DF} &= \overrightarrow{DA} + \overrightarrow{AB} + \overrightarrow{BF} \\ &= -\underline{b} + \underline{a} + \frac{3}{4}\underline{a} \\ \overrightarrow{DE} - \overrightarrow{DF} &= \frac{1}{2}\underline{a}\end{aligned}$$

$$\begin{aligned}\overrightarrow{DE} - \overrightarrow{DF} &= -\underline{b} + \frac{7}{4}\underline{a} - \frac{2}{4}\underline{a} \\ \overrightarrow{FE} &= \frac{5}{4}\underline{a} - \underline{b}\end{aligned}$$

Or other correct approach.

vectors in 2 dimensions

Part A. B & C: Introduction

Unit

selective

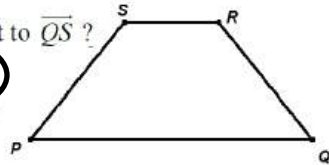
Q17

NEWINGTON

$PQRS$ is a trapezium where $\overrightarrow{PS} = \vec{p}$, $\overrightarrow{SR} = \vec{s}$ and $\overrightarrow{PQ} = 2\overrightarrow{SR}$.

Which of the following is equivalent to $\overrightarrow{QS}$?

- (A) $2\vec{s} + \vec{p}$ (C) $\vec{p} - 2\vec{s}$
(B) $2\vec{s} - \vec{p}$ (D) $\vec{p} - 2\vec{s}$



$$\begin{aligned}\overrightarrow{QS} &= \overrightarrow{PS} - \overrightarrow{PQ} \\ &= \vec{p} - 2\vec{s}\end{aligned}$$

Q18

NEWINGTON

The column (position) vector notation of 4 vectors is shown below.

$$P = \begin{pmatrix} -8 \\ -8 \end{pmatrix} \quad Q = \begin{pmatrix} 3 \\ 6 \end{pmatrix} \quad R = \begin{pmatrix} 1 \\ 5 \end{pmatrix} \quad S = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$$

Find the column (position) vector notation of:

- (i) $\overrightarrow{PQ}$ (ii) $\overrightarrow{RS}$
(iii) $-\overrightarrow{PQ} - \overrightarrow{RS}$

(i) $\overrightarrow{PQ} = \begin{pmatrix} 11 \\ 14 \end{pmatrix}$

(ii) $\overrightarrow{RS} = \begin{pmatrix} -6 \\ 2 \end{pmatrix}$

(iii) $-\overrightarrow{PQ} - \overrightarrow{RS} = \begin{pmatrix} -5 \\ -16 \end{pmatrix}$

Q19

SYDNEY GIRLS

An object moves with velocity 18 m/s at an angle of 150° in the anticlockwise direction to the vector $\vec{i}$. The vector representation of this velocity is:

- A. $-9\sqrt{3}\vec{i} - 9\vec{j}$ C. $-9\sqrt{3}\vec{i} + 9\vec{j}$
B. $-9\vec{i} - 9\sqrt{3}\vec{j}$ D. $-9\vec{i} + 9\sqrt{3}\vec{j}$

Q20

SYDNEY TECH

Which one of the following vectors is parallel to the vector $\overrightarrow{OP} = 12\vec{i} - 6\vec{j}$?

- (A) $\overrightarrow{OA} = 12\vec{i} + 6\vec{j}$ (C) $\overrightarrow{OC} = 2\vec{i} + \vec{j}$
(B) $\overrightarrow{OB} = -\vec{i} + 2\vec{j}$ (D) $\overrightarrow{OD} = -2\vec{i} + \vec{j}$

vectors in 2 dimensions

Part D: Dot Product

3unit

selective

Q 1

ASCHAM

Given $\underline{a} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$

- (i) Find $\underline{a} + \underline{b}$ and $\underline{a} - \underline{b}$ 2
- (ii) Hence, find $(\underline{a} + \underline{b}) \cdot (\underline{a} - \underline{b})$. State if $\underline{a} + \underline{b}$ is perpendicular to $\underline{a} - \underline{b}$ and justify your answer with a reason. 2

(i) $\underline{a} + \underline{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ 7 \end{pmatrix} = \begin{pmatrix} 7 \\ 9 \end{pmatrix} \checkmark$

$\underline{a} - \underline{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 4 \\ 7 \end{pmatrix} = \begin{pmatrix} -1 \\ -5 \end{pmatrix} \checkmark$

(ii) $(\underline{a} + \underline{b}) \cdot (\underline{a} - \underline{b}) = \begin{pmatrix} 7 \\ 9 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -5 \end{pmatrix}$

$= -7 + -45$

$= -52 \checkmark$

$\neq 0$

$\therefore \underline{a} + \underline{b}$ is not perpendicular to $\underline{a} - \underline{b} \checkmark$

Q 2

BAULKHAM

What is the angle between the vectors $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 18 \end{pmatrix}$?

- A. $\theta = \cos^{-1}\left(\frac{39}{65}\right)$
- B. $\theta = \cos^{-1}\left(-\frac{39}{65}\right)$
- C. $\theta = \cos^{-1}\left(\frac{33}{65}\right)$
- D. $\theta = \cos^{-1}\left(-\frac{33}{65}\right)$

$\cos \theta = \frac{-3 \times 1 + 2 \times 18}{\sqrt{(-3)^2 + 2^2} \times \sqrt{1^2 + 18^2}}$

$\cos \theta = \frac{33}{65}$

$\theta = \cos^{-1}\left(\frac{33}{65}\right)$

Q 3

BLACKTOWN

Two vectors are given by $\underline{a} = 3\hat{i} + m\hat{j}$ and $\underline{b} = -10\hat{i} + n\hat{j}$ where $m, n > 0$.

- i) If $|\underline{a}| = 5$ and $\underline{a}$ is perpendicular to $\underline{b}$, find the values of m and n . 2
- ii) Hence find the unit vector $\hat{b}$. 1

$\underline{a} = 3\hat{i} + m\hat{j}$
 $\underline{b} = -10\hat{i} + n\hat{j}$

$|\underline{a}| = 5$
 $3^2 + m^2 = 5^2$
 $m = \sqrt{5^2 - 3^2} \quad (m > 0)$
 $m = 4$

Since $\underline{a}$ is perpendicular to $\underline{b}$
 $\underline{a} \cdot \underline{b} = 0$
 $3 \times -10 + 4 \times n = 0$
 $4n = 30$
 $n = \frac{30}{4}$
 $n = \frac{15}{2}$

$\hat{b} = \frac{\underline{b}}{|\underline{b}|}$

$\hat{b} = \frac{-10\hat{i} + \frac{15}{2}\hat{j}}{\sqrt{(-10)^2 + \left(\frac{15}{2}\right)^2}}$

$\hat{b} = \frac{-10\hat{i} + \frac{15}{2}\hat{j}}{\sqrt{100 + \frac{225}{4}}}$

$\hat{b} = \frac{-10\hat{i} + \frac{15}{2}\hat{j}}{\frac{25}{2}}$

$\hat{b} = \frac{1}{5}(-4\hat{i} + 3\hat{j})$

Q 4

BONNYRIGG

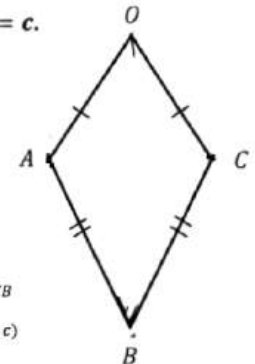
The diagram shows a kite $OABC$. Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ and $\overrightarrow{OC} = \underline{c}$.

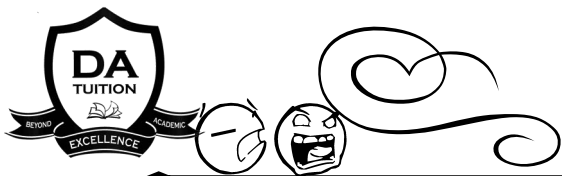
- (i) Show that $(\underline{b} - \underline{a}) \cdot (\underline{b} - \underline{a}) = (\underline{b} - \underline{c}) \cdot (\underline{b} - \underline{c})$ 1
- (ii) Hence prove that the diagonals of a kite are perpendicular by showing that: 1

$(\underline{c} - \underline{a}) \cdot \underline{b} = 0$

(i) $|\underline{b} - \underline{a}|^2 = |\underline{b} - \underline{c}|^2$ since $AB = CB$
 $\therefore (\underline{b} - \underline{a}) \cdot (\underline{b} - \underline{a}) = (\underline{b} - \underline{c}) \cdot (\underline{b} - \underline{c})$

(ii) Expanding the result in (i):
 $|\underline{a}|^2 - 2\underline{a} \cdot \underline{b} + |\underline{b}|^2 = |\underline{b}|^2 - 2\underline{b} \cdot \underline{c} + |\underline{c}|^2$
Since $|\underline{a}| = |\underline{c}|$:
 $2\underline{b} \cdot \underline{c} - 2\underline{a} \cdot \underline{b} = 0 \quad \therefore \underline{b} \cdot (\underline{c} - \underline{a}) = 0$
 $2\underline{b} \cdot (\underline{c} - \underline{a}) = 0$





vectors in 2 dimensions

Part D: Dot Product

Q 5
CARINGBAH

Which of the following vectors is perpendicular to the vector $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$?

(A) $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$

(C) $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$

(B) $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$

(D) $\begin{pmatrix} -2 \\ -4 \end{pmatrix}$

Q 6
CHELTENHAM GIRLS

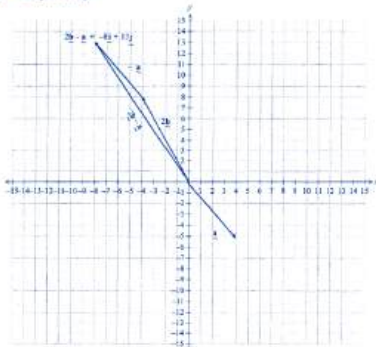
Consider vectors $\underline{a} = 4\mathbf{i} - 5\mathbf{j}$ and $\underline{b} = -2\mathbf{i} + 4\mathbf{j}$.

- Find the magnitude and direction of $\underline{a} + \underline{b}$. 2
- Find the resultant vector of $2\underline{b} - \underline{a}$. 1
- Calculate the dot product $\underline{a} \cdot \underline{b}$. 1
- Find the angle between the vectors $\underline{a}$ and $\underline{b}$. 2

i) $4\mathbf{i} - 5\mathbf{j} + (-2\mathbf{i} + 4\mathbf{j}) = 2\mathbf{i} - \mathbf{j}$
 magnitude of $\underline{a} + \underline{b} = \sqrt{2^2 + (-1)^2} = \sqrt{5}$

direction = α where $\tan \alpha = \left(-\frac{1}{2}\right)$
 $\alpha = \tan^{-1}\left(-\frac{1}{2}\right)$
 $= -26.56505118^\circ$
 $\approx -26^\circ 34'$

ii) Graphically



Algebraically

$2\underline{b} - \underline{a} = 2 \times (-2\mathbf{i} + 4\mathbf{j}) - (4\mathbf{i} - 5\mathbf{j})$
 $= -4\mathbf{i} + 8\mathbf{j} - 4\mathbf{i} + 5\mathbf{j}$
 $= -8\mathbf{i} + 13\mathbf{j}$

iii) $\underline{a} \cdot \underline{b} = x_1 x_2 + y_1 y_2$
 $= 4 \times -2 + -5 \times 4$
 $= -8 - 20$
 $= -28$

iv) $\underline{a} \cdot \underline{b} = |\underline{a}||\underline{b}| \cos \theta$
 where θ is the angle between the vectors.
 $|\underline{a}| = \sqrt{4^2 + (-5)^2} = \sqrt{41}$

$|\underline{b}| = \sqrt{(-2)^2 + 4^2} = \sqrt{20}$
 $\underline{a} \cdot \underline{b} = \sqrt{41} \times \sqrt{20} \cos \theta$
 $\underline{a} \cdot \underline{b} = -28$ (from iii))

$\sqrt{820} \cos \theta = -28$
 $\cos \theta = \frac{-28}{\sqrt{820}}$
 $\theta = \cos^{-1}\left(\frac{-28}{\sqrt{820}}\right)$
 $\theta = \cos^{-1}(-0.9778)$
 $= 167.905^\circ$
 $= 167^\circ 54'$ (nearest minute)

Q 7
CSSA

Find the value of x if the vectors $\underline{a} = 2\mathbf{i} + 3\mathbf{j}$ and $\underline{b} = -3\mathbf{i} + x\mathbf{j}$ are perpendicular.

2

If $\underline{a}$ and $\underline{b}$ are perpendicular then $\underline{a} \cdot \underline{b} = 0$
 $2 \times (-3) + (3) \times x = 0$
 $\therefore x = 2$

Q 8
ENTERPRISE

What is the angle between the vectors $\underline{a} = 2\mathbf{i} - 3\mathbf{j}$ and $\underline{b} = 3\mathbf{i} + \mathbf{j}$

(A) 60.20°

(B) 89.12°

(C) 101.1°

(D) 74.74°

Q 9
FORT STREET

Given the vectors $\underline{a} = 2\mathbf{i} + 3\mathbf{j}$ and $\underline{b} = -3\mathbf{i} - 5\mathbf{j}$,

- Calculate the dot product $\underline{a} \cdot \underline{b}$. 1
- Find $4\underline{a} - 3\underline{b}$ in column vector form. 2

i) $\underline{a} \cdot \underline{b} = (2)(-3) + (3)(-5)$
 $= -21$

ii) $4\underline{a} - 3\underline{b} = 4 \begin{bmatrix} 2 \\ 3 \end{bmatrix} - 3 \begin{bmatrix} -3 \\ -5 \end{bmatrix}$
 $= \begin{bmatrix} 17 \\ 27 \end{bmatrix}$



vectors in 2 dimensions

Part D: Dot Product

3unit

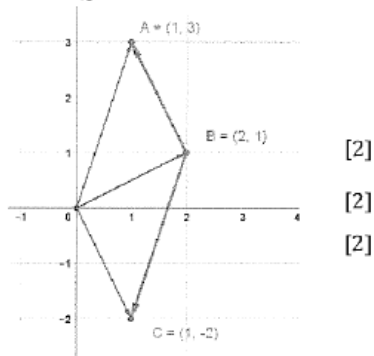
selective

Q10

GIRRAWEEEN

A, B and C are points defined by the position vectors
 $\underline{a} = \underline{i} + 3\underline{j}$, $\underline{b} = 2\underline{i} + \underline{j}$ and $\underline{c} = \underline{i} - 2\underline{j}$ respectively.

- Find $\overrightarrow{BA}$ and $\overrightarrow{BC}$
- Find $|\overrightarrow{BA}|$ and $|\overrightarrow{BC}|$
- Find $\angle ABC$



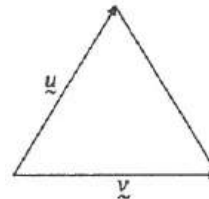
$$\begin{aligned} \text{i) } \overrightarrow{BA} &= \underline{a} - \underline{b} \\ &= (\underline{i} + 3\underline{j}) - (2\underline{i} + \underline{j}) \\ &= -\underline{i} + 2\underline{j} \\ \overrightarrow{BC} &= \underline{c} - \underline{b} \\ &= (\underline{i} - 2\underline{j}) - (2\underline{i} + \underline{j}) \\ &= -\underline{i} - 3\underline{j} \\ \text{ii) } |\overrightarrow{BA}| &= \sqrt{(-1)^2 + 2^2} \\ &= \sqrt{5} \\ |\overrightarrow{BC}| &= \sqrt{(-1)^2 + (-3)^2} \\ &= \sqrt{10} \end{aligned}$$

$$\begin{aligned} \text{iii) } \cos \angle ABC &= \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BA}| \cdot |\overrightarrow{BC}|} \\ &= \frac{(-1 \times -1) + (2 \times -3)}{\sqrt{5} \times \sqrt{10}} \\ &= \frac{-5}{5\sqrt{2}} = \frac{-1}{\sqrt{2}} \\ \angle ABC &= 135^\circ \end{aligned}$$

Q11

GOSFORD

An equilateral triangle of side 3 units is shown below.
Vectors $\underline{u}$ and $\underline{v}$ are represented in the diagram.



What is the value of $\underline{u} \cdot \underline{v}$?

- 0
- $\frac{9}{\sqrt{2}}$
- $\frac{9}{2}$
- 9

Q12

GOSFORD

Let $\underline{u} = \underline{i} + \underline{j}$ and $\underline{v} = \underline{i} - \underline{j}$. What is the angle between the two vectors.

- $\frac{\pi}{2}$
- $\frac{\pi}{4}$
- π
- 2π

Q13

GOSFORD

Points A, B have position vectors $\overrightarrow{OA} = 3\underline{i} - 2\underline{j}$, $\overrightarrow{OB} = -\underline{i} + \underline{j}$

- Find the unit vector along $\overrightarrow{AB}$
- Suppose P is a point on AB such that $\overrightarrow{OP} \perp \overrightarrow{AB}$, Find $\overrightarrow{OP}$

$$\begin{aligned} \text{c) } \underline{a} &= \begin{pmatrix} 3 \\ -2 \end{pmatrix} \quad \underline{b} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \\ \overrightarrow{AB} &= \underline{a} - \underline{b} \\ &= \begin{pmatrix} 3 \\ -2 \end{pmatrix} - \begin{pmatrix} -1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ -3 \end{pmatrix} \\ |\overrightarrow{AB}| &= \sqrt{4^2 + (-3)^2} = 5 \\ \hat{AB} &= \frac{1}{5} \begin{pmatrix} 4 \\ -3 \end{pmatrix} \\ \text{ii) } \overrightarrow{OP} &= \lambda \overrightarrow{AB} \\ &= \lambda \begin{pmatrix} 4 \\ -3 \end{pmatrix} \\ \overrightarrow{OP} \cdot \overrightarrow{AB} &= 0 \quad \overrightarrow{OP} \perp \overrightarrow{AB} \\ \begin{pmatrix} 4\lambda \\ -3\lambda \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -3 \end{pmatrix} &= 0 \\ 16\lambda + 9\lambda &= 0 \\ 25\lambda &= 0 \\ \lambda &= 0 \\ \overrightarrow{OP} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{aligned}$$

vectors in 2 dimensions

Part D: Dot Product

Unit

Selective

Q14

GRAFTON

Explain why $\hat{i} \cdot \hat{j} = 0$

1

$\hat{i}$ and $\hat{j}$ are defined as perpendicular unit vectors.
 $\therefore \hat{i} \cdot \hat{j} = |\hat{i}| |\hat{j}| \cos 90$
 $= 1 \times 1 \times 0$
 $= 0$

Q15

HUNTERS HILL

Which of the following vectors is perpendicular to $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$?

(A) $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$

(C) $\begin{bmatrix} -6 \\ -4 \end{bmatrix}$

(B) $\begin{bmatrix} -3 \\ 2 \end{bmatrix}$

(D) $\begin{bmatrix} -4 \\ 6 \end{bmatrix}$

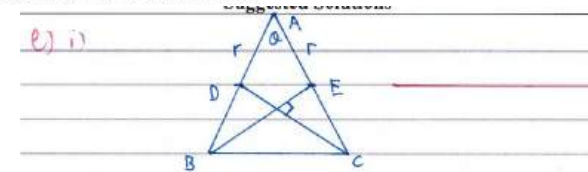
If $\begin{bmatrix} a \\ b \end{bmatrix}$ is perpendicular,
 then $-2a + 3b = 0$
 $\frac{a}{b} = \frac{3}{2}$
 $\begin{bmatrix} -6 \\ -4 \end{bmatrix}$ satisfies this

Q16

JAMES RUSE

In the isosceles triangle ABC , $|\overline{AB}| = |\overline{AC}|$. D is the midpoint of side AB and E is the midpoint of side AC . $\overline{CD}$ is perpendicular to $\overline{BE}$.

- Draw the diagram and label $\angle BAC = \theta$ and $|\overline{AD}| = r$. 1
- Hence noting that $\overline{CD}$ may be written as $\overline{AD} - \overline{AC}$, or otherwise, use vector methods to find the value of $\angle BAC$. 3



i) $\overline{CD} = \overline{AD} - \overline{AC}$
 $\overline{BE} = \overline{AE} - \overline{AB}$

$(\overline{CD}) \cdot (\overline{BE}) = 0$ (Perpendicular vectors)

$(\overline{AD} - \overline{AC}) \cdot (\overline{AE} - \overline{AB}) = 0$

$\overline{AD} \cdot \overline{AE} - \overline{AD} \cdot \overline{AB} - \overline{AC} \cdot \overline{AE} + \overline{AC} \cdot \overline{AB} = 0$

$\overline{AD} \cdot \overline{AE} + \overline{AC} \cdot \overline{AB} = \overline{AD} \cdot \overline{AB} + \overline{AC} \cdot \overline{AE}$

$\overline{AD} \cdot \overline{AE} + \overline{AC} \cdot \overline{AB} = \overline{AD} \cdot 2\overline{AD} + 2\overline{AE} \cdot \overline{AE}$
 $= 2r^2 + 2r^2$
 $= 4r^2$

$|\overline{AB}| |\overline{AE}| \cos \theta + |\overline{AC}| |\overline{AB}| \cos \theta = 4r^2$

$r^2 \cos \theta + 4r^2 \cos \theta = 4r^2$

$\cos \theta (5r^2) = 4r^2$

$\cos \theta = \frac{4}{5}$

$\theta = \cos^{-1} \frac{4}{5}$

$\approx 36^\circ 52'$

vectors in 2 dimensions

Part D: Dot Product

3unit

selective

Q17

MANLY

A, B and C are points defined by the position vectors $\underline{a} = \underline{i} + 3\underline{j}$, $\underline{b} = 2\underline{i} + \underline{j}$ and $\underline{c} = \underline{i} - 2\underline{j}$ respectively.

- i) Find $\overrightarrow{AB} \cdot \overrightarrow{CB}$ 1
- ii) Hence, using part i., find the size of $\angle ABC$. 2

c-i

$$\begin{aligned}\overrightarrow{AB} &= (2\underline{i} + \underline{j}) - (\underline{i} + 3\underline{j}) \\ &= \underline{i} - 2\underline{j} \\ \overrightarrow{CB} &= (2\underline{i} + \underline{j}) - (\underline{i} - 2\underline{j}) \\ &= \underline{i} + 3\underline{j} \\ |\overrightarrow{AB}| &= \sqrt{5} \quad |\overrightarrow{CB}| = \sqrt{10} \\ \overrightarrow{AB} \cdot \overrightarrow{CB} &= 1 \times 1 + (-2 \times 3) = -5\end{aligned}$$

c-ii

$$\begin{aligned}\cos \theta &= \frac{m \cdot n}{|m| |n|} \\ &= \frac{-5}{\sqrt{5} \sqrt{10}} \\ \cos \theta &= -\frac{1}{\sqrt{2}} \\ \theta &= \frac{3\pi}{4}\end{aligned}$$

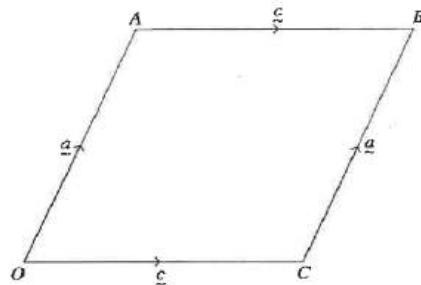
Q18

NEAP

The diagram shows $OABC$, a rhombus in which $\overrightarrow{OA} = \overrightarrow{CB} = \underline{a}$ and $\overrightarrow{OC} = \overrightarrow{AB} = \underline{c}$.

To prove that the diagonals of $OABC$ are perpendicular, it is required to show that

- (A) $(\underline{a} + \underline{c}) \cdot (\underline{a} + \underline{c}) = 0$.
- (B) $(\underline{a} - \underline{c}) \cdot (\underline{a} - \underline{c}) = 0$.
- (C) $(\underline{a} - \underline{c}) \cdot (\underline{a} + \underline{c}) = 0$.
- (D) $\underline{a} \cdot \underline{c} = 0$.



The diagonals of $OABC$ are given by $\overrightarrow{OB}$ and $\overrightarrow{CA}$.

To prove they are perpendicular, form $\overrightarrow{CA} \cdot \overrightarrow{OB}$ and show that it equals zero.

$$\overrightarrow{CA} = \underline{a} - \underline{c} \text{ and } \overrightarrow{OB} = \underline{a} + \underline{c}.$$

Therefore $\overrightarrow{CA} \cdot \overrightarrow{OB} = 0$ if $(\underline{a} - \underline{c}) \cdot (\underline{a} + \underline{c}) = 0$.

Q19

GIRRAWEEEN

A vector perpendicular to $3\underline{i} + 4\underline{j}$ and with magnitude 5 is

- A. $-4\underline{i} - 3\underline{j}$ C. $4\underline{i} - 3\underline{j}$
- B. $5(4\underline{i} - 3\underline{j})$ D. $\sqrt{5}(4\underline{i} - 3\underline{j})$

$$\begin{aligned}(\underline{4i} - \underline{3j}) \cdot (3\underline{i} + 4\underline{j}) \\ &= (4 \times 3 - 3 \times 4) = 0 \Rightarrow \text{perpendicular} \\ |\underline{4i} - \underline{3j}| &= \sqrt{4^2 + (-3)^2} = 5 \Rightarrow \text{magnitude 5}\end{aligned}$$

vectors in 2 dimensions

Part D: Dot Product

3unit

selective

Q20

NORTH SYD GIRLS

Consider the vectors $\underline{u} = 2\underline{i} - 5\underline{j}$, $\underline{v} = \underline{i} + a\underline{j}$ and $\underline{w} = \underline{i} + b\underline{j}$.

If $\underline{u} \parallel \underline{v}$ and $\underline{u} \perp \underline{w}$, what are possible values for a and b ?

A. $a = \frac{5}{2}, b = \frac{2}{5}$

C. $a = -\frac{5}{2}, b = \frac{2}{5}$

B. $a = \frac{5}{2}, b = -\frac{2}{5}$

D. $a = -\frac{5}{2}, b = -\frac{2}{5}$

$$\underline{u} \cdot \underline{w} = 0$$

$$(2\underline{i} - 5\underline{j}) \cdot (\underline{i} + b\underline{j}) = 0$$

$$2 - 5b = 0$$

$$b = \frac{2}{5}$$

$$\underline{v} = \frac{1}{2}\underline{u}$$

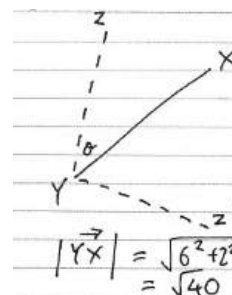
$$\therefore a = -\frac{5}{2}$$

Q21

ST GEORGE GIRLS

An acute-angled triangle XYZ has an area of 40 square units.

The vector $\overrightarrow{YX} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$ and $\overrightarrow{YZ} = \begin{bmatrix} p \\ q \end{bmatrix}$. Given $|\overrightarrow{YZ}| = 8\sqrt{5}$, find the possible values of p and q .



$$A = \frac{1}{2} \times \sqrt{40} \times 8\sqrt{5} \times \sin \theta$$

$$40 = 4\sqrt{200} \times \sin \theta$$

$$\sin \theta = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ \text{ (since } \theta \text{ is acute)}$$

$$\therefore \begin{bmatrix} 6 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} p \\ q \end{bmatrix} = \sqrt{40} \times 8\sqrt{5} \times \cos 45^\circ$$

$$6p + 2q = 80\sqrt{2} \times \frac{1}{\sqrt{2}} \quad (1)$$

$$6p + 2q = 80$$

$$3p + q = 40$$

$$\therefore q = 40 - 3p \quad (1)$$

Also, given $|\overrightarrow{YZ}| = 8\sqrt{5}$

$$\sqrt{p^2 + q^2} = 8\sqrt{5}$$

$$p^2 + q^2 = 320 \quad (2)$$

Sub (1) into (2)

$$p^2 + (40 - 3p)^2 = 320 \quad (1)$$

$$p^2 + 1600 - 240p + 9p^2 = 320$$

$$10p^2 - 240p + 1280 = 0$$

$$p^2 - 24p + 128 = 0$$

$$(p - 8)(p - 16) = 0$$

$$\therefore p = 8, p = 16$$

when $p = 8$, $q = 40 - 3(8)$
 $= 16$

when $p = 16$, $q = 40 - 3(16)$
 $= -8$

$$\therefore \begin{bmatrix} p \\ q \end{bmatrix} = \begin{bmatrix} 8 \\ 16 \end{bmatrix} \text{ or } \begin{bmatrix} 16 \\ -8 \end{bmatrix} \quad (1)$$

vectors in 2 dimensions

Part D: Dot Product

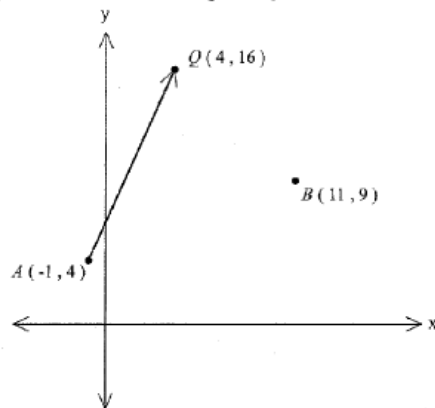
Unit

Selective

Q22

SYDNEY TECH

Adam walks in a straight line from the point $A(-1, 4)$ to the point $Q(4, 16)$ with constant speed. His position vector can be expressed in the form $\underline{p} = \underline{a} + t\underline{u}$, where t is the time after he starts walking. Adam arrives at the point Q at $t = 3$.



- State the vectors $\underline{a}$ and $\underline{u}$. 2
- Bob is at point $B(11, 9)$. During Adam's walk from A to Q , Bob wishes to throw a ball to Adam. Bob decides to throw the ball when Adam is at the closest point to B .
 - Write a vector $\underline{w}$ that is perpendicular to $\underline{u}$. 1
 - Hence or otherwise, find value of t when Bob throws the ball to Adam. 3

$$b) \therefore \underline{AQ} = \begin{bmatrix} 4 - (-1) \\ 16 - 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 12 \end{bmatrix}$$

and Adam gets to Q from A at $t = 3$,

$$\text{Hence } \underline{a} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} \text{ and } \underline{u} = \frac{1}{3} \times \begin{bmatrix} 5 \\ 12 \end{bmatrix} = \begin{bmatrix} \frac{5}{3} \\ 4 \end{bmatrix}$$

ii/ a) A vector perpendicular is

$$\underline{w} = \begin{bmatrix} -4 \\ \frac{5}{3} \end{bmatrix} \text{ or any scalar multiple.}$$

b) Equation of the throw is: $\begin{bmatrix} 11 \\ 9 \end{bmatrix} + \lambda \begin{bmatrix} -4 \\ \frac{5}{3} \end{bmatrix}$, as it goes from $(11, 9)$ in the direction $\underline{w}$.

$$\text{Hence } \begin{bmatrix} -1 \\ 4 \end{bmatrix} + t \begin{bmatrix} \frac{5}{3} \\ 4 \end{bmatrix} = \begin{bmatrix} 11 \\ 9 \end{bmatrix} + \lambda \begin{bmatrix} -4 \\ \frac{5}{3} \end{bmatrix}$$

$$\begin{bmatrix} -1 + \frac{5t}{3} \\ 4 + 4t \end{bmatrix} = \begin{bmatrix} 11 - 4\lambda \\ 9 + \frac{5\lambda}{3} \end{bmatrix}$$

$$\therefore -1 + \frac{5t}{3} = 11 - 4\lambda \quad \text{and} \quad 4 + 4t = 9 + \frac{5\lambda}{3}$$

$$-3 + 5t = 33 - 12\lambda$$

$$12\lambda = 36 - 5t$$

$$\lambda = \frac{36 - 5t}{12} \quad (1)$$

Sub (1) into (2)

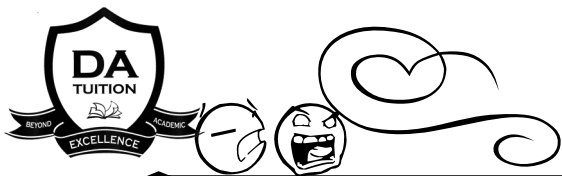
$$12 + 12t = 27 + 5 \times \frac{36 - 5t}{12}$$

$$75 + 12t = 180 - 25t$$

$$169t = 360$$

$$t = \frac{360}{169} \quad (\text{or } t = 2.13 \text{ (2dp)})$$





vectors in 2 dimensions

Part E: Projections

3unit

selective

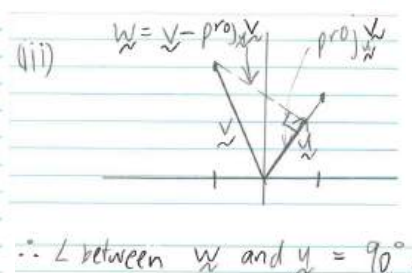
Q 1

ASCHAM

Given $\underline{u} = 2\hat{i} + 3\hat{j}$ and $\underline{v} = -2\hat{i} + 4\hat{j}$,

- Find $\text{proj}_{\underline{u}} \underline{v}$ 2
- Find $\underline{w} = \underline{v} - \text{proj}_{\underline{u}} \underline{v}$ 1
- What is the angle between $\underline{u}$ and $\underline{w}$? 1

$$\begin{aligned} \text{i) } \text{proj}_{\underline{u}} \underline{v} &= \frac{\underline{v} \cdot \underline{u}}{\underline{u} \cdot \underline{u}} \underline{u} & \underline{u} &= 2\hat{i} + 3\hat{j} \\ &= \frac{(-2 \times 2 + 3 \times 4)}{(2^2 + 3^2)} (2\hat{i} + 3\hat{j}) & \underline{v} &= -2\hat{i} + 4\hat{j} \\ &= \frac{8}{13} (2\hat{i} + 3\hat{j}) \\ &= \frac{16}{13}\hat{i} + \frac{24}{13}\hat{j} \quad \checkmark \\ \text{ii) } \underline{w} &= \underline{v} - \text{proj}_{\underline{u}} \underline{v} = -2\hat{i} + 4\hat{j} - \left(\frac{16}{13}\hat{i} + \frac{24}{13}\hat{j}\right) \\ &= -\frac{42}{13}\hat{i} + \frac{28}{13}\hat{j} \end{aligned}$$



Q 2

BAULKHAM

If $\underline{u}$ and $\underline{v}$ are two non-zero vectors on the Cartesian plane, which of the following statements is **NOT ALWAYS CORRECT**?

- $|\text{proj}_{\underline{v}} \underline{u}| \leq |\underline{u}|$
- $|\text{proj}_{\underline{u}} \underline{v}| \leq |\underline{v}|$
- $|\text{proj}_{\underline{v}} \underline{u}| = \frac{|\underline{u} \cdot \underline{v}|}{|\underline{v}|}$
- $|\text{proj}_{\underline{v}} \underline{u}| = |\underline{u} \cdot \hat{\underline{v}}|$

By the geometry of projection of $\underline{u}$ onto $\underline{v}$, its length must be shorter than or equal to the length of $\underline{u}$.

The "signed" length of $\text{proj}_{\underline{v}} \underline{u}$ is $\underline{u} \cdot \hat{\underline{v}}$, therefore the length of $\text{proj}_{\underline{v}} \underline{u}$ is $|\underline{u} \cdot \hat{\underline{v}}|$.

Similarly, the "signed" length of $\text{proj}_{\underline{u}} \underline{v}$ is $\underline{v} \cdot \hat{\underline{u}} = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}|}$, therefore the length of $\text{proj}_{\underline{u}} \underline{v}$ is $\left| \frac{\underline{u} \cdot \underline{v}}{|\underline{u}|} \right| = \frac{|\underline{u} \cdot \underline{v}|}{|\underline{u}|}$

A counter example of option B could be $\underline{u} = 2\hat{i} + 2\hat{j}$ and $\underline{v} = \hat{j}$.

Q 3

BONNYRIGG

Let U, V and W be the points corresponding to the position vectors $\underline{u} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, $\underline{v} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$

and $\underline{w} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}$ respectively. Find:

- $3\underline{v} - 2\underline{u}$ 1
- $\text{proj}_{\underline{u}} \underline{v}$ 2
- $\angle UVW$ (leave your answer to the nearest degree) 2

$$\text{(i) } 3\underline{v} - 2\underline{u} = \begin{pmatrix} -6 \\ 12 \end{pmatrix} - \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} -12 \\ 10 \end{pmatrix}$$

$$\text{(ii) } \text{proj}_{\underline{u}} \underline{v} = \frac{\underline{v} \cdot \underline{u}}{|\underline{u}|^2} \underline{u} = -\frac{2}{10} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = -\frac{1}{5} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\text{(iii) } \underline{VU} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}$$

$$\underline{VW} = \begin{pmatrix} 7 \\ -8 \end{pmatrix}$$

$$\begin{aligned} \angle UVW &= \cos^{-1} \left(\frac{\underline{VU} \cdot \underline{VW}}{|\underline{VU}| |\underline{VW}|} \right) \\ &= \cos^{-1} \left(\frac{59}{\sqrt{34} \cdot \sqrt{113}} \right) \\ &= 17^\circ 51' = 18^\circ \text{ (nearest degree)} \end{aligned}$$

Q 4

CHELTENHAM GIRLS

Find the vector projection of $\underline{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ onto $\underline{q} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

- A. $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$ C. $\begin{pmatrix} -2\sqrt{5} \\ 4\sqrt{5} \end{pmatrix}$
 B. $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$ D. $\begin{pmatrix} 2\sqrt{5} \\ -4\sqrt{5} \end{pmatrix}$

$$\text{Proj}_{\underline{q}}(\underline{p}) = \frac{\underline{p} \cdot \underline{q}}{|\underline{q}|^2} \cdot \underline{q}$$

$$\underline{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \text{ onto } \underline{q} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$$\underline{p} \cdot \underline{q} = 4 \times -1 + -3 \times 2 = -10$$

$$|\underline{q}|^2 = (-1)^2 + 2^2 = 5$$

$$\text{Proj}_{\underline{q}}(\underline{p}) = \frac{-10}{5} \cdot \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$$= -2 \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ -4 \end{pmatrix}$$

Q 5

CSSA

Given the vectors $\underline{a} = 2\underline{i} - 5\underline{j}$ and $\underline{b} = 3\underline{i} + 4\underline{j}$, what is $\text{proj}_{\underline{a}} \underline{b}$?

- (A) $\frac{14}{29}(2\underline{i} - 5\underline{j})$ (C) $\frac{-14}{29}(2\underline{i} - 5\underline{j})$
 (B) $\frac{14}{\sqrt{29}}(2\underline{i} - 5\underline{j})$ (D) $\frac{-14}{\sqrt{29}}(2\underline{i} - 5\underline{j})$

$$\underline{a} = 2\underline{i} - 5\underline{j}, \underline{b} = 3\underline{i} + 4\underline{j}$$

$$\text{proj}_{\underline{a}} \underline{b} = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}|^2} \underline{a}$$

$$= \frac{2 \times 3 + (-5 \times 4)}{(\sqrt{2^2 + 5^2})^2} [2\underline{i} - 5\underline{j}]$$

$$= \frac{-14}{29} [2\underline{i} - 5\underline{j}]$$

Hence (C)

Q 6

ENTERPRISE

Given $\underline{c} = 2\underline{i} - 8\underline{j}$

- (i) find $\hat{\underline{c}}$ 2
 (ii) find vector $\underline{d}$ of magnitude 17 in the direction of $\underline{c}$ 1

$$(i) |\underline{c}| = \sqrt{2^2 + (-8)^2} = \sqrt{68} = 2\sqrt{17}$$

$$\hat{\underline{c}} = \frac{1}{2\sqrt{17}} (2\underline{i} - 8\underline{j}) = \frac{1}{\sqrt{17}} (\underline{i} - 4\underline{j})$$

$$(ii) \underline{d} = \frac{17}{\sqrt{17}} (\underline{i} - 4\underline{j}) = \sqrt{17} (\underline{i} - 4\underline{j})$$

Q 7

ENTERPRISE

Consider the vectors $\underline{a} = 2\underline{i} - 4\underline{j}$ and $\underline{b} = -3\underline{i} + 5\underline{j}$

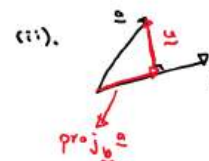
- (i) Find the vector projection of $\underline{a}$ onto $\underline{b}$ 3
 (ii) Find the vector projection of $\underline{a}$ perpendicular to $\underline{b}$ 2

$$(i), \text{proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|^2} \underline{b}$$

$$= \frac{2 \times -3 + (-4) \times 5}{(-3)^2 + 5^2} \underline{b}$$

$$= -\frac{26}{34} \underline{b}$$

$$= -\frac{13}{17} (-3\underline{i} + 5\underline{j})$$

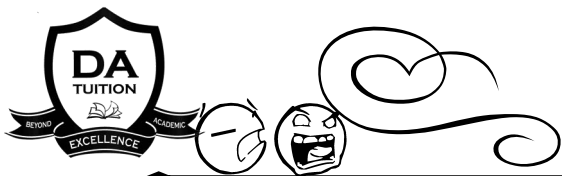


$$(ii), \underline{u} = \underline{a} - \text{proj}_{\underline{b}} \underline{a}$$

$$= (2\underline{i} - 4\underline{j}) - \left(-\frac{13}{17}(-3\underline{i} + 5\underline{j})\right)$$

$$= \left(2 + \frac{13}{17} \times 3\right) \underline{i} + \left(-4 + \frac{13}{17} \times 5\right) \underline{j}$$

$$= -\frac{5}{17} \underline{i} - \frac{3}{17} \underline{j}$$



vectors in 2 dimensions

Part e: Projections

3unit

selective

Q 8

GRAFTON

Consider the vectors $\underline{a} = 2\hat{i} + 3\hat{j}$ and $\underline{b} = 9\hat{i} + \hat{j}$

(i) Calculate the angle, in radians to 3 decimal places, between the vectors $\underline{a}$ and $\underline{b}$ 2

(ii) Find the projection of $\underline{a}$ onto $\underline{b}$ 2

$$\begin{aligned} \text{i) } \cos \theta &= \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|} & \underline{a} &= 2\hat{i} + 3\hat{j} \\ & & \underline{b} &= 9\hat{i} + \hat{j} \\ \therefore \cos \theta &= \frac{(2 \times 9) + (3 \times 1)}{\sqrt{2^2 + 3^2} \times \sqrt{9^2 + 1^2}} & & \\ &= \frac{21}{\sqrt{13} \sqrt{82}} & & \\ &= 0.64319... & & \\ \theta &= 0.872136... & & \\ &= 0.872 (3d.p.) \text{ in radians } \textcircled{2} \end{aligned}$$

Q 9

JAMES RUSE

(i) Given $A(-2, 3)$, $B(4, -5)$, $C(-7, -6)$ and $D(-5, -2)$, find the vector projection of $\overline{AB}$ on to $\overline{CD}$. 2

(ii) What is the vector component of $\overline{AB}$ perpendicular to $\overline{CD}$? 1

$$\begin{aligned} \text{i) } \underline{AB} &= \begin{pmatrix} 4 \\ -5 \end{pmatrix} - \begin{pmatrix} -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ -8 \end{pmatrix} & |\underline{AB}| &= \sqrt{6^2 + (-8)^2} = 10 \\ \underline{CD} &= \begin{pmatrix} -5 \\ -2 \end{pmatrix} - \begin{pmatrix} -7 \\ -6 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} & |\underline{CD}| &= \sqrt{2^2 + 4^2} = 2\sqrt{5} \\ \text{proj}_{\underline{CD}} \underline{AB} &= \frac{\underline{AB} \cdot \underline{CD}}{|\underline{CD}|^2} \times \underline{CD} & & \\ &= \frac{\begin{pmatrix} 6 \\ -8 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 4 \end{pmatrix}}{(2\sqrt{5})^2} \times \begin{pmatrix} 2 \\ 4 \end{pmatrix} & & \\ &= \frac{(6 \times 2) + (-8 \times 4)}{20} \times \begin{pmatrix} 2 \\ 4 \end{pmatrix} & & \\ &= \frac{-20}{20} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ -4 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{ii) } \text{perp. } \underline{AB} &= \text{perp. from pt (i)} \\ &= \begin{pmatrix} 6 \\ -8 \end{pmatrix} - \begin{pmatrix} -2 \\ -4 \end{pmatrix} \\ &= \begin{pmatrix} 8 \\ -4 \end{pmatrix} \end{aligned}$$

Q10

KILLARA

Consider vectors $\underline{a} = 4\hat{i} - 5\hat{j}$ and $\underline{b} = -2\hat{i} + 4\hat{j}$.

(i) Find the magnitude and direction of $\underline{a} + \underline{b}$. 2

(ii) Calculate the dot product $\underline{a} \cdot \underline{b}$. 1

(iii) Find the projection of $\underline{a}$ in the direction of $\underline{b}$. 2

$$\begin{aligned} \text{i) } \underline{a} &= 4\hat{i} - 5\hat{j} \text{ \& } \underline{b} = -2\hat{i} + 4\hat{j} \\ \text{ii) } \underline{a} \cdot \underline{b} &= 4\hat{i} \cdot (-2\hat{i}) + (-5\hat{j}) \cdot 4\hat{j} = -8 - 20 = -28 \\ \text{iii) } \text{proj}_{\underline{b}} \underline{a} &= \frac{(\underline{a} \cdot \underline{b}) \underline{b}}{|\underline{b}|^2} = \frac{-28}{16} (-2\hat{i} + 4\hat{j}) \end{aligned}$$

Q11

MANLY

Find $\text{Proj}_{\underline{w}} \underline{v}$ given $\underline{v} = -2\hat{i} - 5\hat{j}$ and $\underline{w} = 3\hat{i} + \hat{j}$

- A. $-\frac{11}{10} (3\hat{i} + \hat{j})$ C. $-\frac{11}{10} (-2\hat{i} - 5\hat{j})$
B. $-\frac{11}{29} (3\hat{i} + \hat{j})$ D. $-\frac{11}{29} (-2\hat{i} - 5\hat{j})$

$\text{Proj}_{\underline{w}} \underline{v}$ given $\underline{v} = -2\hat{i} - 5\hat{j}$ and $\underline{w} = 3\hat{i} + \hat{j}$

$$\begin{aligned} \underline{v} \cdot \underline{w} &= -2 \times 3 - 5 \times 1 = -11 \\ |\underline{v}| &= \sqrt{(-2)^2 + (-5)^2} = \sqrt{29} \\ |\underline{w}| &= \sqrt{3^2 + 1^2} = \sqrt{10} \end{aligned}$$

$$\begin{aligned} \text{Proj}_{\underline{w}} \underline{v} &= \frac{\underline{v} \cdot \underline{w}}{|\underline{w}|^2} \underline{w} \\ &= \frac{-11}{10} (3\hat{i} + \hat{j}) \end{aligned}$$

vectors in 2 dimensions

Part E: Projections

3unit

selective

Q12

PRAIRIEWOOD

A particle is moving with velocity $\underline{v} = (15\mathbf{i} + 20\mathbf{j})\text{ms}^{-1}$.

- Find the unit vector in the direction of the vector $4\mathbf{i} + 3\mathbf{j}$.
- Find the component of $\underline{v}$ in the direction of the vector $4\mathbf{i} + 3\mathbf{j}$.

(i) Unit Vector:

$$\sqrt{4^2 + 3^2} = 5$$

Hence required unit vector is $\frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j}$

(ii) $(15\mathbf{i} + 20\mathbf{j}) \cdot (\frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j})$

$$= 12 + 12$$

$$= 24$$

Required component of $\underline{v}$ is $24(\frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j})$

$$= \frac{96}{5}\mathbf{i} + \frac{72}{5}\mathbf{j}$$

Q13

PRAIRIEWOOD

Given $\underline{a} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$ and $\underline{b} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$, which of the following represents $\text{proj}_{\underline{b}} \underline{a}$?

- (A) $\begin{bmatrix} 40 \\ -20 \end{bmatrix}$ (C) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$
- (B) $\begin{bmatrix} 20 \\ -10 \end{bmatrix}$ (D) $\begin{bmatrix} \frac{4}{5} \\ -\frac{2}{5} \end{bmatrix}$

$$\text{proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{\underline{b} \cdot \underline{b}} \underline{b}$$

$$= \frac{20 - 10}{4 + 4} \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

$$= \frac{10}{8} \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ -2.5 \end{bmatrix}$$

Q14

NORMANHURST

A force, described by the vector $\underline{F} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, moves a particle along the line ℓ from $A(-1, -1)$ to $B(2, -3)$.

- Find the unit vector that is in the direction of $\overrightarrow{AB}$. 2
- Find the component of $\underline{F}$ in the direction of line ℓ . 2

i. $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ ii. Applying the projection of $\underline{F}$ on to $\overrightarrow{AB}$, such that

$$|\overrightarrow{AB}| = \sqrt{9 + 4} = \sqrt{13}$$

$$\therefore \text{unit vector is } \frac{1}{\sqrt{13}} \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

$$\text{proj}_{\overrightarrow{AB}} \underline{F} = \frac{\underline{F} \cdot \overrightarrow{AB}}{|\overrightarrow{AB}|^2} \overrightarrow{AB}$$

$$= \frac{\begin{pmatrix} 2 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \end{pmatrix}}{13} \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

$$= \frac{6 - 10}{13} \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

$$= -\frac{4}{13} \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

Q15

NORTH SYD BOYS

- Given $\underline{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ find the vector projection of $\underline{a}$ onto $\underline{b}$. 2
- Calculate the angle between the vectors $\underline{a}$ and $\underline{b}$. 1

a) i) $\text{proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|^2} \underline{b}$ ii) $\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta$

$$= \frac{\begin{pmatrix} 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \end{pmatrix}}{\sqrt{4+1}} \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$= \frac{-1}{\sqrt{5}} \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$= \begin{bmatrix} -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

$\frac{-1}{\sqrt{10} \sqrt{5}} = \cos \theta$

$$\theta = 98.69^\circ$$

vectors in 2 dimensions

Part E: Projections

3unit

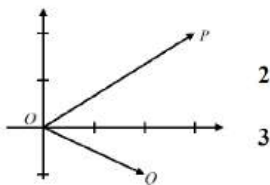
selective

Q16

NORTH SYDN GIRLS

Consider the vectors $u = \overrightarrow{OP} = 3\mathbf{i} + 2\mathbf{j}$ and $v = \overrightarrow{OQ} = 2\mathbf{i} - \mathbf{j}$.

- Find the vector projection $\overrightarrow{OR}$ of v onto u .
- Use vector methods to find the exact area of $\triangle OPQ$.



$$\begin{aligned} \text{i) } \text{proj}_u(v) &= \frac{u \cdot v}{u \cdot u} u \\ &= \frac{(3)(2) + (2)(-1)}{(3)^2 + (2)^2} (3\mathbf{i} + 2\mathbf{j}) \\ &= \frac{4}{13} (3\mathbf{i} + 2\mathbf{j}) \\ \overrightarrow{OR} &= \frac{12}{13}\mathbf{i} + \frac{8}{13}\mathbf{j} \end{aligned}$$

[2]

$$\begin{aligned} \text{ii) } \overrightarrow{RQ} &= \overrightarrow{OQ} - \overrightarrow{OR} \\ &= (2\mathbf{i} - \mathbf{j}) - \left(\frac{12}{13}\mathbf{i} + \frac{8}{13}\mathbf{j}\right) \\ &= \frac{10}{13}\mathbf{i} - \frac{21}{13}\mathbf{j} \\ &= 7\left(\frac{10}{91}\mathbf{i} - \frac{21}{91}\mathbf{j}\right) \Rightarrow |\overrightarrow{RQ}| = 7 \cdot \sqrt{\left(\frac{10}{91}\right)^2 + \left(\frac{21}{91}\right)^2} \\ &= \frac{7}{\sqrt{13}} \end{aligned}$$

$$|\overrightarrow{OP}| = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$\text{Area} = \frac{1}{2} \times \sqrt{13} \times \frac{7}{\sqrt{13}} = \frac{7}{2} \text{ units}^2$$

[3]

Q17

PENRITH

What is the length of the vector projection of $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ onto $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$?

A. $\frac{3\sqrt{13}}{13}$ units

C. $\frac{3\sqrt{10}}{13}$ units

B. $\frac{3\sqrt{10}}{10}$ units

D. $\frac{3\sqrt{13}}{10}$ units

$$\begin{aligned} |\text{proj}_b a| &= \left| \frac{a \cdot b}{b \cdot b} \right| |b| \\ &= \frac{6-3}{4+9} \left| \begin{pmatrix} 2 \\ -3 \end{pmatrix} \right| \\ &= \frac{3}{13} |\sqrt{13}| \\ &= \frac{3\sqrt{13}}{13} \end{aligned}$$

[A]

Q18

ST GEORGE GIRLS

Consider the points $P(a, 2a)$, $Q(-a, 5a)$, $R(3a, 4a)$ and $S(9a, 12a)$, where a is a positive real number.

- Express $\overrightarrow{PQ}$ in component form.
- Given the length of the projection of $\overrightarrow{PQ}$ onto $\overrightarrow{RS}$ is 12, find the value of a .

1

3

$$\begin{aligned} \text{b) i) } \overrightarrow{PQ} &= (-a-a)\mathbf{i} + (5a-2a)\mathbf{j} \\ &= -2a\mathbf{i} + 3a\mathbf{j} \\ &\text{or } (-2a, 3a) \end{aligned}$$

Method 1 (easiest way to do)

ii) Scalar projection of $\overrightarrow{PQ}$ onto $\overrightarrow{RS}$

$$= \frac{\overrightarrow{PQ} \cdot \overrightarrow{RS}}{|\overrightarrow{RS}|}$$

$$\begin{aligned} \overrightarrow{RS} &= (9a-3a)\mathbf{i} + (12a-4a)\mathbf{j} \\ &= 6a\mathbf{i} + 8a\mathbf{j} \end{aligned}$$

Dot product of $\overrightarrow{PQ} \cdot \overrightarrow{RS}$

$$\begin{aligned} \overrightarrow{PQ} &= (-2a, 3a) \quad \overrightarrow{RS} = (6a, 8a) \\ \overrightarrow{PQ} \cdot \overrightarrow{RS} &= -2a \times 6a + 3a \times 8a \\ &= -12a^2 + 24a^2 \end{aligned}$$

$$\begin{aligned} |\overrightarrow{RS}| &= \sqrt{(6a)^2 + (8a)^2} \\ &= \sqrt{36a^2 + 64a^2} \\ &= \sqrt{100a^2} \end{aligned}$$

$$\therefore \frac{\overrightarrow{PQ} \cdot \overrightarrow{RS}}{|\overrightarrow{RS}|} = \frac{-12a^2 + 24a^2}{10a} = 12$$

$$\frac{12a^2}{10a} = 12$$

$$\frac{6a}{5} = 12$$

$$a = 10$$

vectors in 2 dimensions

PART F: PROOF USING VECTORS

3unit

selective

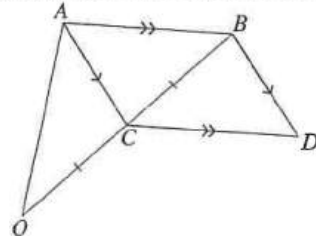
Q 1

NEAP

The position vectors of points A and B are $\underline{a}$ and $\underline{b}$ respectively. Point C is the midpoint of OB and point D is such that $ABDC$ is a parallelogram.

Which of the following is the position vector of D ?

- (A) $\frac{3}{2}\underline{b} + \underline{a}$ (C) $\frac{1}{2}\underline{b} - \frac{1}{2}\underline{a}$
(B) $\frac{3}{2}\underline{b} - \underline{a}$ (D) $\frac{1}{2}\underline{b} - \underline{a}$

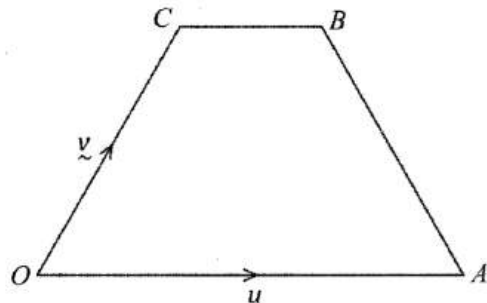


$$\begin{aligned}\vec{OD} &= \vec{OC} + \vec{CD} \\ &= \frac{1}{2}\vec{OB} + \vec{AB} \\ &= \frac{1}{2}\vec{OB} + \vec{AO} + \vec{OB} \\ &= \frac{1}{2}\underline{b} - \underline{a} + \underline{b} \\ &= \frac{3}{2}\underline{b} - \underline{a}\end{aligned}$$

Q 2

SYDNEY TECH

In the trapezium below, $\vec{OA} = \underline{u}$, $\vec{OC} = \underline{v}$, and $|\vec{OA}| = 2|\vec{CB}|$



Express the vector $\vec{BA}$ in terms of $\underline{u}$ and $\underline{v}$.

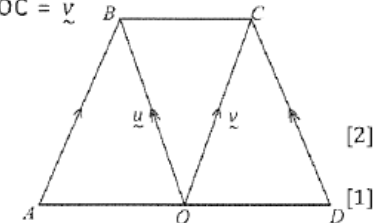
$$\begin{aligned}\vec{BA} &= \frac{1}{2}\underline{u} - \underline{v} + \underline{u} \\ &= \frac{3}{2}\underline{u} - \underline{v}\end{aligned}$$

Q 3

GIRRAWEE

$AB \parallel OC$; $DC \parallel OB$; $\vec{OB} = \underline{u}$; $\vec{OC} = \underline{v}$
 $AB = OB = OC = DC$

- (i) Express $\vec{AD}$ in terms of $\underline{u}$ and $\underline{v}$
(ii) Express $\vec{BD}$ in terms of $\underline{u}$ and $\underline{v}$

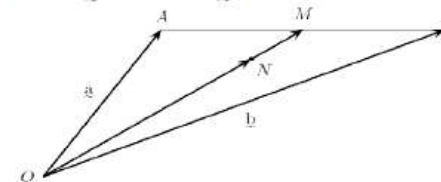


$$\begin{aligned}\text{i) } \vec{AD} &= \vec{AB} - \vec{OB} \\ &= \underline{v} - \underline{u} \\ \vec{AD} &= \vec{OD} - \vec{OA} \quad (\triangle ABD \cong \triangle OCD) \\ \vec{AD} &= \vec{AO} + \vec{OD} \\ &= (\underline{v} - \underline{u}) + (\underline{v} - \underline{u}) \\ &= 2\underline{v} - 2\underline{u} \\ \text{ii) } \vec{BD} &= \vec{AD} - \vec{AB} \\ &= (2\underline{v} - 2\underline{u}) - \underline{v} \\ &= \underline{v} - 2\underline{u}\end{aligned}$$

Q 4

NORMANHURST

In the following diagram, M is the midpoint of line segment AB , and $\vec{ON} = \frac{4}{5}\vec{OM}$.
It is given that $\vec{OA} = \underline{a}$ and $\vec{OB} = \underline{b}$.



Which of the following expressions correctly relates $\vec{NM}$ with $\underline{a}$ and $\underline{b}$?

- (A) $\vec{NM} = \frac{4}{5}(\underline{a} - \underline{b})$ (C) $\vec{NM} = \frac{1}{10}(\underline{a} + \underline{b})$
(B) $\vec{NM} = \frac{2}{5}(\underline{a} + \underline{b})$ (D) $\vec{NM} = \frac{1}{10}(\underline{a} - \underline{b})$

vectors in 2 dimensions

Part F: Proof using vectors

3unit

selective

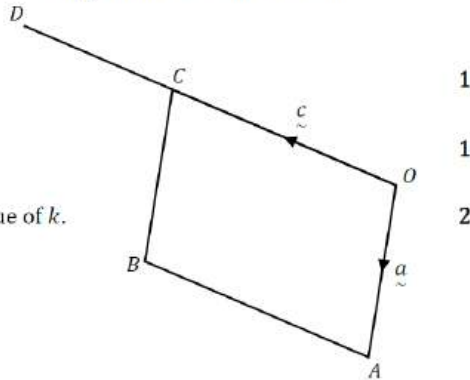
Q 5

HUNTERS HILL

In the parallelogram $OABC$, $\vec{OA} = \underline{a}$ and $\vec{OC} = \underline{c}$.

X is the midpoint of the line AC , and OCD is a straight line such that the ratio $OC : CD = k : 1$.

- Find $\vec{OX}$ in terms of $\underline{a}$ and $\underline{c}$.
- Write an expression for $\vec{OD}$.
- Given $\vec{XD} = 3\underline{c} - \frac{1}{2}\underline{a}$, find the value of k .



1

1

2

$$\begin{aligned} \text{i) } \vec{OX} &= \frac{1}{2}\underline{c} + \frac{1}{2}\underline{a} \\ &= \frac{1}{2}(\underline{a} + \underline{c}) \end{aligned}$$

$$= \frac{k+2}{2k}\underline{c} - \frac{1}{2}\underline{a}$$

$$\text{ii) } \vec{OD} = \frac{k+1}{k}\vec{OC} = \frac{k+1}{k}\underline{c}$$

$$\text{as } \frac{k+2}{2k}\underline{c} - \frac{1}{2}\underline{a} = 3\underline{c} - \frac{1}{2}\underline{a}$$

$$\begin{aligned} \text{iii) } \vec{XD} &= \vec{OD} - \vec{OX} \\ &= \frac{k+1}{k}\underline{c} - \frac{1}{2}(\underline{a} + \underline{c}) \\ &= \left(\frac{k+1}{k} - \frac{1}{2}\right)\underline{c} - \frac{1}{2}\underline{a} \\ &= \frac{2k+2-k}{2k}\underline{c} - \frac{1}{2}\underline{a} \end{aligned}$$

$$\frac{k+2}{2k} = 3 \quad (\text{equating } \underline{c} \text{ component})$$

$$k+2 = 6k$$

$$5k = 2$$

$$k = \frac{2}{5}$$

Q 6

NEAP

$ABCD$ is a rhombus with $\vec{AB} = \underline{a}$ and $\vec{AD} = \underline{d}$.

Use vector methods to prove that the diagonals of the rhombus are perpendicular to each other.

$$\vec{AC} = \underline{a} + \underline{d}, \vec{DB} = \underline{a} - \underline{d}$$

$$\vec{AC} \cdot \vec{DB} = (\underline{a} + \underline{d}) \cdot (\underline{a} - \underline{d})$$

$$= \underline{a} \cdot \underline{a} - \underline{d} \cdot \underline{d}$$

$$= |\underline{a}|^2 - |\underline{d}|^2$$

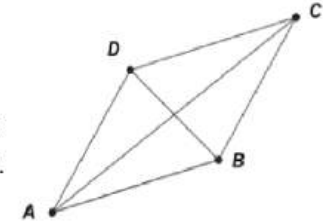
$$= 0 \quad (\text{sides of rhombus have equal length})$$

But $\vec{AC} \cdot \vec{DB} = |\underline{a}| |\underline{d}| \cos \theta$ (where θ is the angle between the vectors)

$$\therefore \cos \theta = 0$$

$$\theta = 90^\circ$$

i.e. diagonals are perpendicular



Q 7

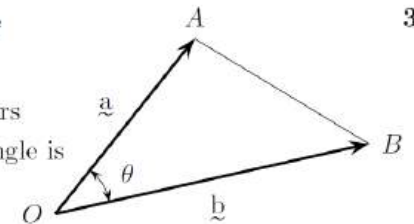
NORMANHURST

The diagram below shows $\triangle OAB$, where

$\vec{OA} = \underline{a}$ and $\vec{OB} = \underline{b}$.

Given that θ is the angle between vectors $\underline{a}$ and $\underline{b}$, show that the area of the triangle is

$$A = \frac{1}{2} \sqrt{(\underline{a} \cdot \underline{a})(\underline{b} \cdot \underline{b}) - (\underline{a} \cdot \underline{b})^2}$$



3

$$A = \frac{1}{2} |\underline{a}| |\underline{b}| \sin \theta$$

$$A^2 = \frac{1}{4} |\underline{a}|^2 |\underline{b}|^2 \left(1 - \frac{(\underline{a} \cdot \underline{b})^2}{|\underline{a}|^2 |\underline{b}|^2} \right)$$

$$A^2 = \frac{1}{4} |\underline{a}|^2 |\underline{b}|^2 \sin^2 \theta$$

$$= \frac{1}{4} (|\underline{a}|^2 |\underline{b}|^2 - (\underline{a} \cdot \underline{b})^2)$$

$$= \frac{1}{4} |\underline{a}|^2 |\underline{b}|^2 (1 - \cos^2 \theta)$$

$$= \frac{1}{4} ((\underline{a} \cdot \underline{a})(\underline{b} \cdot \underline{b}) - (\underline{a} \cdot \underline{b})^2)$$

$$\therefore \cos \theta = \frac{(\underline{a} \cdot \underline{b})}{|\underline{a}| |\underline{b}|}$$

$$\therefore A = \frac{1}{2} \sqrt{(\underline{a} \cdot \underline{a})(\underline{b} \cdot \underline{b}) - (\underline{a} \cdot \underline{b})^2}$$

$$\cos^2 \theta = \frac{(\underline{a} \cdot \underline{b})^2}{|\underline{a}|^2 |\underline{b}|^2}$$

vectors in 2 dimensions

Part F: Proof using vectors

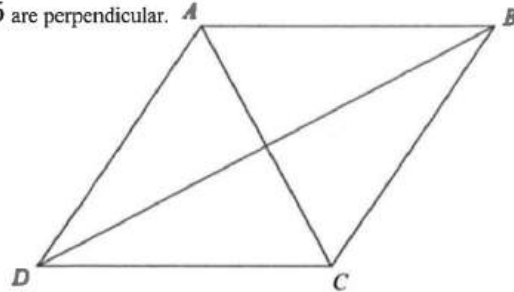
3unit

selective

Q 8

CSSA

In the diagram $ABCD$ is a parallelogram. If $AB = BC$, prove that the vectors $\overrightarrow{AC}$ and $\overrightarrow{BD}$ are perpendicular.



let $\overrightarrow{AB} = \underline{a}$ and $\overrightarrow{BC} = \underline{b}$
since $ABCD$ is a parallelogram
 $\overrightarrow{CD} = -\underline{a}$ and $\overrightarrow{DA} = -\underline{b}$

$\therefore \overrightarrow{AC} = \underline{a} + \underline{b}$ and $\overrightarrow{DB} = \underline{a} - \underline{b}$

to prove AC and DB are perpendicular we show $\overrightarrow{AC} \cdot \overrightarrow{DB} = 0$

$$(\underline{a} + \underline{b}) \cdot (\underline{a} - \underline{b}) = |\underline{a}|^2 - |\underline{b}|^2$$

$$= 0$$

since $AB = BC$

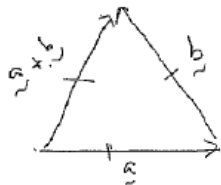
$\therefore \overrightarrow{AC}$ and $\overrightarrow{DB}$ are perpendicular

Q 9

CSSA

If the sum of two unit vectors is a unit vector, ($\underline{a}$, $\underline{b}$ and $\underline{a} + \underline{b}$ are all unit vectors) prove using vector properties that the magnitude of their difference

$$|\underline{a} - \underline{b}| = \sqrt{3}$$



Consider

$$\begin{aligned} \underline{a} \cdot \underline{b} &= |\underline{a}| |\underline{b}| \cos \frac{2\pi}{3} \\ &= 1 \times 1 \times \frac{-1}{2} = -\frac{1}{2} \end{aligned}$$

Consider

$$\begin{aligned} (\underline{a} - \underline{b}) \cdot (\underline{a} - \underline{b}) &= |\underline{a} - \underline{b}|^2 \\ |\underline{a} - \underline{b}|^2 &= \underline{a} \cdot \underline{a} - \underline{a} \cdot \underline{b} - \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b} \\ &= |\underline{a}|^2 + |\underline{b}|^2 - 2\underline{a} \cdot \underline{b} \\ &= 1 + 1 - 2 \times \left(-\frac{1}{2}\right) = 3 \\ |\underline{a} - \underline{b}| &= \sqrt{3} \end{aligned}$$

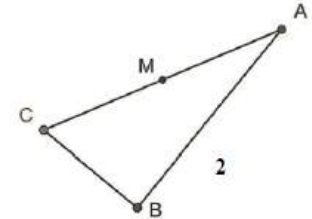
Q10

FORT STREET

$\triangle ABC$ is a right-angled with M being the midpoint of the hypotenuse AC , as shown.

Let $\overrightarrow{AM} = \underline{a}$ and $\overrightarrow{BM} = \underline{b}$.

- Find $\overrightarrow{AB}$ and $\overrightarrow{BC}$ in terms of $\underline{a}$ and $\underline{b}$.
- Prove that M is equidistant from the three vertices of $\triangle ABC$



$$\begin{aligned} \text{i) } \overrightarrow{AB} &= \overrightarrow{AM} + \overrightarrow{MB} & \overrightarrow{BC} &= \overrightarrow{BM} + \overrightarrow{MC} \\ &= \underline{a} - \underline{b} & &= \underline{b} + \underline{a} \\ &= \underline{a} - \underline{b} & &= \underline{a} + \underline{b} \end{aligned}$$

$$\begin{aligned} \text{ii) } \overrightarrow{AB} \cdot \overrightarrow{BC} &= (\underline{a} - \underline{b}) \cdot (\underline{a} + \underline{b}) \\ &= \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} - \underline{a} \cdot \underline{b} - \underline{b} \cdot \underline{b} \\ &= |\underline{a}|^2 + 0 + |\underline{b}|^2 \\ &= |\underline{a}|^2 + |\underline{b}|^2 \end{aligned}$$

Since $\triangle ABC$ is a right-angled triangle and $\overrightarrow{AB} \perp \overrightarrow{BC}$.

$$\text{then } \overrightarrow{AB} \cdot \overrightarrow{BC} = 0 \Rightarrow |\underline{a}|^2 - |\underline{b}|^2 = 0$$

$$\therefore |\underline{a}| = |\underline{b}|$$

$$\therefore |\overrightarrow{AM}| = |\overrightarrow{BM}| = |\overrightarrow{MC}| \text{ as required.}$$

vectors in 2 dimensions

Part F: Proof using vectors

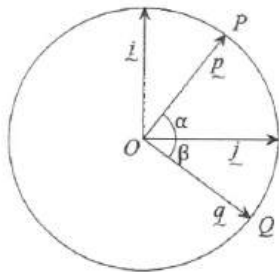
Unit

Selective

Q11

GOSFORD

Use the unit circle and the vectors in the diagram to derive the expansion of $\cos(\alpha + \beta)$



3

A man throws a ball from a point O on the horizontal ground so that it lands on the ground at a point P distant 80 m from him. The ball reaches the highest point 20 m above the ground. Neglect the sizes of the man and the ball. Take $g = 10 \text{ m/s}^2$

$$\cos(\alpha + \beta) = \frac{p \cdot q}{|p||q|}$$

$$|p| = |q| = 1$$

$$\cos(\alpha + \beta) = p \cdot q$$

$$p = \cos \alpha i + \sin \alpha j$$

$$q = \cos \beta i - \sin \beta j$$

$$p \cdot q = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

Q12

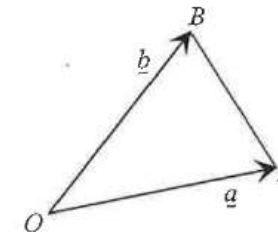
INDEPENDENT

In the diagram, OAB is an acute angled triangle in which $OA = OB$. The vector $\vec{OA} = \underline{a}$ and $\vec{OB} = \underline{b}$.

(i) Show that $\cos \angle OAB = \frac{\underline{a} \cdot \underline{a} - \underline{a} \cdot \underline{b}}{|\underline{a}| \times |\underline{a} - \underline{b}|}$

and $\cos \angle OBA = \frac{\underline{b} \cdot \underline{b} - \underline{a} \cdot \underline{b}}{|\underline{a}| \times |\underline{a} - \underline{b}|}$

(ii) Hence show that $\angle OAB = \angle OBA$.



$$\begin{aligned} \text{i. } \cos \angle OAB &= \frac{\vec{AO} \cdot \vec{AB}}{AO \times AB} \\ &= \frac{(-\underline{a}) \cdot (\underline{b} - \underline{a})}{|\underline{a}| \times |\underline{b} - \underline{a}|} \\ &= \frac{\underline{a} \cdot \underline{a} - \underline{a} \cdot \underline{b}}{|\underline{a}| \times |\underline{a} - \underline{b}|} \end{aligned}$$

$$\begin{aligned} \cos \angle OBA &= \frac{\vec{BO} \cdot \vec{BA}}{BO \times BA} \\ &= \frac{\underline{b} \cdot \underline{b} - \underline{a} \cdot \underline{b}}{|\underline{b}| \times |\underline{a} - \underline{b}|} \quad (\text{interchanging } \underline{a} \leftrightarrow \underline{b}) \\ &= \frac{\underline{b} \cdot \underline{b} - \underline{a} \cdot \underline{b}}{|\underline{a}| \times |\underline{a} - \underline{b}|} \quad (\text{since } OA = OB \Rightarrow |\underline{a}| = |\underline{b}|) \end{aligned}$$

ii. $\underline{a} \cdot \underline{a} = |\underline{a}|^2 = |\underline{b}|^2 = \underline{b} \cdot \underline{b}$ (since $OA = OB \Rightarrow |\underline{a}| = |\underline{b}|$)
 $\therefore \cos \angle OAB = \cos \angle OBA$. Hence since both angles are acute, $\angle OAB = \angle OBA$.

vectors in 2 dimensions

Part F: Proof using vectors

3unit

selective

Q13

NORTH SYD BOYS

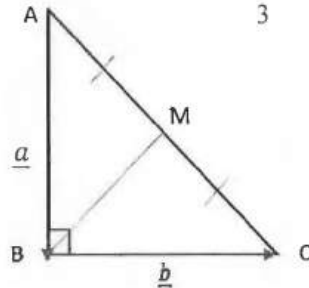
Consider the right angled triangle ABC below.

You are given $|\underline{b} + \underline{a}|^2 = |\underline{b}|^2 + 2\underline{a} \cdot \underline{b} + |\underline{a}|^2$

and $|\underline{b} - \underline{a}|^2 = |\underline{b}|^2 - 2\underline{a} \cdot \underline{b} + |\underline{a}|^2$

You are not required to prove these.

Hence, or otherwise, use vector methods to prove that the midpoint M of the hypotenuse of the right angled triangle is equidistant from all three vertices.



$$\begin{aligned} \text{b) } \underline{AC} &= \underline{a} + \underline{b} \\ \underline{AM} &= \frac{1}{2} \underline{AC} \\ &= \frac{1}{2} (\underline{a} + \underline{b}) \\ \text{Sim, } \underline{MC} &= \frac{1}{2} (\underline{a} + \underline{b}) \\ \underline{BM} &= \underline{BC} + \underline{CM} \\ &= \underline{b} - \frac{1}{2} (\underline{a} + \underline{b}) \\ &= \frac{1}{2} (\underline{b} - \underline{a}) \\ |\underline{AM}|^2 &= \frac{1}{4} (\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b}) \\ &= \frac{1}{4} (\underline{a} \cdot \underline{a} + 2\underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{b}) \\ &= \frac{1}{4} |\underline{a} + \underline{b}|^2 \end{aligned}$$

$$\begin{aligned} \text{Sim, } |\underline{MC}|^2 &= \frac{1}{4} |\underline{a} + \underline{b}|^2 \\ &= \frac{1}{4} |\underline{AC}|^2 \end{aligned}$$

$$\begin{aligned} |\underline{BM}|^2 &= \frac{1}{4} [(\underline{b} - \underline{a}) \cdot (\underline{b} - \underline{a})] \\ &= \frac{1}{4} (\underline{b} \cdot \underline{b} - 2\underline{a} \cdot \underline{b} + \underline{a} \cdot \underline{a}) \\ &= \frac{1}{4} |\underline{b} - \underline{a}|^2 \\ &= \frac{1}{4} |\underline{AC}|^2 \quad |\underline{b} + \underline{a}|^2 = |\underline{b} - \underline{a}|^2 \text{ as } 2\underline{a} \cdot \underline{b} = 0 \\ &\quad \text{as } \underline{a} \perp \underline{b} \end{aligned}$$

$$\therefore |\underline{BM}| = |\underline{AM}| = |\underline{MC}|$$

Q14

PENRITH

Use vectors to prove that the sum of the squares of the lengths of the two diagonals of a parallelogram is equal to the sum of the squares of the lengths of the four sides.

b)

OACB is a parallelogram

Let $\underline{OA} = \underline{a}$, $\underline{OB} = \underline{b}$

$$\begin{aligned} \underline{OC} &= \underline{a} + \underline{b} \\ \underline{BA} &= \underline{a} - \underline{b} \\ |\underline{OC}|^2 + |\underline{BA}|^2 &= |\underline{a} + \underline{b}|^2 + |\underline{a} - \underline{b}|^2 \\ &= (\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b}) + (\underline{a} - \underline{b}) \cdot (\underline{a} - \underline{b}) \\ &= \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b} \\ &\quad + \underline{a} \cdot \underline{a} - \underline{a} \cdot \underline{b} - \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b} \\ &= 2|\underline{a}|^2 + 2|\underline{b}|^2 \\ &= |\underline{a}|^2 + |\underline{b}|^2 + |\underline{a}|^2 + |\underline{b}|^2 \end{aligned}$$

note that parallelogram has two pairs of parallel & equal sides.

3 marks

vectors in 2 dimensions

Part F: Proof using vectors

3unit

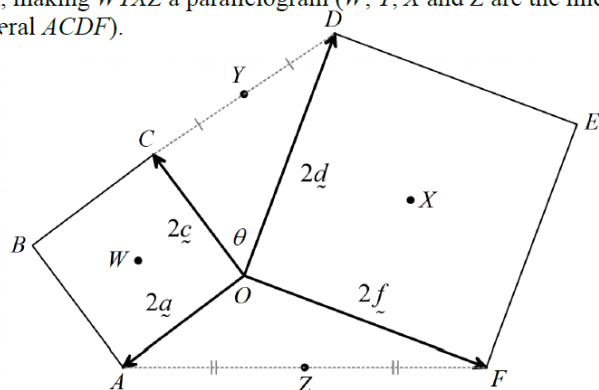
selective

Q15

BAULKHAM

In the diagram, $OABC$ and $ODEF$ are two non-overlapping squares. W and X are the centres of these two squares respectively.

Points Y and Z are the midpoints on the intervals CD and AF respectively, as shown in the diagram, making $WYXZ$ a parallelogram (W , Y , X and Z are the midpoints of the sides of quadrilateral $ACDF$).



Let A , C , D and F have position vectors $2\mathbf{a}$, $2\mathbf{c}$, $2\mathbf{d}$ and $2\mathbf{f}$ respectively.

It can be shown that the interval joining the midpoint of two sides of a triangle is parallel to the third side and is half the length of this third side. You may use this result without proving it.

- (i) Consider $\triangle ACF$, express $\overrightarrow{WZ}$ in terms of $\mathbf{c}$ and $\mathbf{f}$. Similarly, find an expression for $\overrightarrow{WY}$ in terms of $\mathbf{a}$ and $\mathbf{d}$ using $\triangle ACD$. 2
- (ii) $\triangle ADO$ and $\triangle CFO$ are congruent triangles. By considering the lengths of AD and CF , show that $|\overrightarrow{WY}| = |\overrightarrow{WZ}|$. 1
- (iii) Let $\angle COD = \theta$ as shown above, show that $\mathbf{a} \cdot \mathbf{f} = -\mathbf{c} \cdot \mathbf{d}$. 2
- (iv) By considering the dot product of vectors $\overrightarrow{WY}$ and $\overrightarrow{WZ}$, as well as the results from parts (ii) and (iii), show that $WYXZ$ is a square. 2

$$\begin{aligned} \text{(i)} \quad \overrightarrow{WZ} &= \frac{1}{2} \overrightarrow{CF} \\ &= \frac{1}{2} (\overrightarrow{OF} - \overrightarrow{OC}) \\ &= \frac{1}{2} (2\mathbf{f} - 2\mathbf{c}) \\ &= \mathbf{f} - \mathbf{c} \end{aligned} \quad \begin{aligned} \overrightarrow{WY} &= \frac{1}{2} \overrightarrow{AD} \\ &= \frac{1}{2} (\overrightarrow{OD} - \overrightarrow{OA}) \\ &= \frac{1}{2} (2\mathbf{d} - 2\mathbf{a}) \\ &= \mathbf{d} - \mathbf{a} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad |\overrightarrow{AD}| &= |\overrightarrow{CF}| \\ \frac{1}{2} |\overrightarrow{AD}| &= \frac{1}{2} |\overrightarrow{CF}| \\ |\overrightarrow{WY}| &= |\overrightarrow{WZ}| \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \angle AOF &= 180^\circ - \theta \\ \overrightarrow{OA} \cdot \overrightarrow{OF} &= |\overrightarrow{OA}| |\overrightarrow{OF}| \cos(\angle AOF) \\ (2\mathbf{a}) \cdot (2\mathbf{f}) &= |2\mathbf{a}| |2\mathbf{f}| \cos(180^\circ - \theta) \\ 4\mathbf{a} \cdot \mathbf{f} &= -|2\mathbf{a}| |2\mathbf{f}| \cos \theta \quad (\text{since } \cos(180^\circ - \theta) = -\cos \theta) \\ 4\mathbf{a} \cdot \mathbf{f} &= -|2\mathbf{c}| |2\mathbf{d}| \cos \theta \quad (\text{since } |2\mathbf{a}| = |2\mathbf{c}| \text{ and } |2\mathbf{f}| = |2\mathbf{d}|) \\ 4\mathbf{a} \cdot \mathbf{f} &= -(\overrightarrow{OC} \cdot \overrightarrow{OD}) \\ 4\mathbf{a} \cdot \mathbf{f} &= -2\mathbf{c} \cdot 2\mathbf{d} \\ \mathbf{a} \cdot \mathbf{f} &= -\mathbf{c} \cdot \mathbf{d} \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad \overrightarrow{WY} \cdot \overrightarrow{WZ} &= (\mathbf{d} - \mathbf{a}) \cdot (\mathbf{f} - \mathbf{c}) \\ &= \mathbf{d} \cdot \mathbf{f} - \mathbf{d} \cdot \mathbf{c} - \mathbf{a} \cdot \mathbf{f} + \mathbf{a} \cdot \mathbf{c} \\ &= \mathbf{d} \cdot \mathbf{f} - \mathbf{d} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{d} + \mathbf{a} \cdot \mathbf{c} \quad (\text{from part (iii)}) \\ &= \mathbf{d} \cdot \mathbf{f} + \mathbf{a} \cdot \mathbf{c} \\ &= \frac{1}{4} (2\mathbf{d} \cdot 2\mathbf{f}) + \frac{1}{4} (2\mathbf{a} \cdot 2\mathbf{c}) \\ &= 0 \quad (\text{since } 2\mathbf{d} \cdot 2\mathbf{f} = 0 \text{ as } OD \perp OF \text{ and } 2\mathbf{a} \cdot 2\mathbf{c} = 0 \text{ as } OA \perp OC) \end{aligned}$$

Since $|\overrightarrow{WY}| = |\overrightarrow{WZ}|$ and $\overrightarrow{WY} \cdot \overrightarrow{WZ} = 0$ (i.e. $WY \perp WZ$), parallelogram $WYXZ$ is a square (a pair of adjacent sides equal and perpendicular in a parallelogram).



vectors in 2 dimensions

Part F: Proof using vectors

3unit

selective

Q16

SYDNEY GRAMMAR

Points $P(4, -6)$, $Q(1, 2)$, $R(7, 5)$ form a triangle PQR in the Cartesian Plane.

- Find the vectors $\overrightarrow{QP}$ and $\overrightarrow{QR}$, representing two sides of this triangle. Give your answer in component form. 1
- Use the dot product to find angle PQR . Give your answer correct to the nearest degree. 2

c $P(4, -6)$ $Q(1, 2)$ $R(7, 5)$

$$\begin{aligned}\overrightarrow{QP} &= P - Q \\ &= \begin{pmatrix} 4-1 \\ -6-2 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ -8 \end{pmatrix} \\ &= 3\mathbf{i} - 8\mathbf{j}\end{aligned}\quad \begin{aligned}\overrightarrow{QR} &= R - Q \\ &= \begin{pmatrix} 7-1 \\ 5-2 \end{pmatrix} \\ &= \begin{pmatrix} 6 \\ 3 \end{pmatrix} \\ &= 6\mathbf{i} + 3\mathbf{j}\end{aligned}$$

$$\begin{aligned}\overrightarrow{QP} \cdot \overrightarrow{QR} &= 3 \times 6 + (-8) \times 3 \\ &= 18 - 24 \\ &= -6 \\ &= |\overrightarrow{QP}| |\overrightarrow{QR}| \cos \angle PQR\end{aligned}$$

$$\begin{aligned}|\overrightarrow{QP}| &= \sqrt{3^2 + 8^2} \\ &= \sqrt{73} \\ |\overrightarrow{QR}| &= \sqrt{6^2 + 3^2} \\ &= \sqrt{45} \\ &= 3\sqrt{5}\end{aligned}$$

$$\begin{aligned}\therefore \cos \angle PQR &= \frac{-6}{3\sqrt{5}\sqrt{73}} \\ &= \frac{-2}{\sqrt{365}} \\ \angle PQR &= \cos^{-1} \frac{-2}{\sqrt{365}} \\ &\doteq 96^\circ \text{ (nearest degree)}\end{aligned}$$